

FISHERIES

Last 5 Previous Year Question

- 1. What is the Rearing period for fresh water prawns in monoculture?**
 - a. 7 to 10 months
 - b. 6 to 8 months
 - c. 10 to 12 months
 - d. 4 to 5 months

- 2. Which of the following fresh water fishes has a high demand in market?**
 - a. Rohu
 - b. mullets
 - c. Pearlsompt
 - d. Milkfish

- 3. Blue Revolution is a term used for increased production of**
 - a. Petroleum products
 - b. Flower production
 - c. Fish production
 - d. None of the above

- 4. In Inland the cultivation of fish is banned in India because of its predatory nature and it can affect the ecological balance**
 - a. Magur fish
 - b. Migral fish
 - c. Bata fish
 - d. All of the above

- 5. Which of the following indigenous fish is suitable to culture with bottom and column feeders in a polyculture system?**
 - a. Milkfish
 - b. Cuttlefish
 - c. Labeo rohita
 - d. None of the above

- 6. What should be the optimum level of Dissolve oxygen in fish pond?**
- Less than 5 mg/lit
 - More than 10 mg/lit
 - More then 5 mg/lit
 - Less than 10 mg/lit
- 7. When the pH of the fish tank levels become too high or too low, there may be a chance of alkaline or acidic death of the fish. What is the Optimum pH level for most freshwater fish?**
- 6.5 to 7.5
 - 5.5 to 4.5
 - 7.5 to 8.5
 - Neutral pH at 7
- 8. Exotic fish variety that is world's important cultured fish, now well acclimated in Indian waters.**
- White carp
 - Catla catla
 - Bata Labeo bata
 - None of the above
- 9. Nodal agency for the holistic development of seafood industry in India to realize its full export potential?**
- Dept of export and import
 - MPEDA
 - CIBA
 - None of the above

Answers

- | | | | | | | |
|-------------|------------|------------|------------|------------|------------|------------|
| 1. B | 2.A | 3.C | 4.A | 5.C | 6.C | 7.A |
| 8.A | 9.B | | | | | |

FISHERIES

Fisheries

Fisheries is an economic activity that **involves harvesting fish or any aquatic organism from the wild (Capture Fisheries) or raising them in confinement (Culture Fisheries/ Aquaculture)**. It may be Traditional/ Small Scale Fisheries (SSF) for sustenance, or Large-Scale/ Commercial Fisheries for profit.

Fisheries Can Be Categorised into Two Types.

1. **Fin Fisheries** - The **fin fisheries of true fishes**, Term used to describe **the strictly classified biological group of fishes, sometimes called true fishes** to distinguish them from other aquatic life whose common **names also end in “fish”**, including mollusks (e.g., cuttlefish), crustaceans (e.g., crayfish), echinoderms (e.g., starfish), and other animals (e.g., jellyfish); or any other aquatic life harvested in fisheries or aquaculture (e.g. shellfish).



Figure 1 – Fin Fisheries

2. **Non-Fin Fisheries.** - The **non-fin fisheries is the fisheries of organisms other than true fish like prawn, crab, lobster, mussel, oyster, sea cucumbers, frog, sea weeds, etc.**



Figure 2 – Non-Fin Fisheries (Prawn)

Fin Fisheries Can Be Further Categorised into Two Types

1. **Capture Fisheries-** Capture fisheries are **exploitation of aquatic organisms without stocking the seed**. Recruitment of the **species occur naturally**. This is carried out in **the sea, rivers, reservoirs, etc.**

- **Fish yield decreases gradually in capture fisheries** due to indiscriminate catching of fish including brooders and juveniles.
- **Overfishing** destroys the fish stocks.
- **Pollution and environmental factors** influence the fish yield. The catches include both desirable and undesirable varieties.



Figure 3 – Capture Fisheries

2. Culture Fisheries

Culture fisheries are the cultivation of selected fishes in confined areas with utmost care to get maximum yield.

- The **seed is stocked, nursed and reared in confined waters**, then the crop is harvested. Culture takes place in ponds, which are fertilized and supplementary feeds are provided to fish to get maximum yield.
- In order **to overcome the problems found in capture fisheries to increase the production**, considerable attention is being given to the culture fisheries.
- Culture fisheries is **conducted in freshwater, brackish water and sea waters**.
- With the development and expansion of **new culture systems, farming of a wide variety of aquatic organisms like prawns, crabs, molluscs, frogs, sea weeds, etc. have come under culture fisheries**.
- **Due to the culture of a variety of aquatic organisms, culture fisheries have been termed as aquaculture.**



Figure 4 – Culture Fisheries

Aquaculture

Aquaculture production has increased tremendously during the last decade. Consequently, **the percentage share of aquaculture in total inland fish production is estimated to be about 75-80%**. Two specific aqua-produced, **carps and prawns in freshwater aquaculture and shrimps in brackish water aquaculture, have contributed to the bulk as well as value of the inland aquaculture sector.**

There are basically five types of aquacultural practices followed in India they are

1. Freshwater aquaculture,
2. Brackish water aquaculture,
3. Mariculture,
4. Coldwater aquaculture
5. Ornamental fish culture.

1. Freshwater Aquaculture

Freshwater Fish/Aquaculture: Fish that spend **most or all of their life in freshwaters**, such as **rivers and lakes, Streams, Estuaries and Backwaters** having a **salinity of less than 0.5 ppt**. Around 40% of all known species of fish are found in freshwater.

They may be divided into

- **Coldwater Fish (5 – 20 °C)**; examples: Mahseer, Trout, etc.,
- **Warmwater Fish (25 – 35 °C)**; example: Carps, Catfish, Snakeheads, Featherbacks, etc.

Around 70% of India's fish production comes from inland waters, of which nearly 65% comes from aquaculture. Out of the total inland aquaculture production, **Indian Major Carps are the most cultured freshwater fish followed by Exotic Carps, Minor Carps, Catfish and Trout.**

There **are around 1300 Carp Hatcheries in India** that produced seed and supply to fish farmers.

Various fish species in Fresh water (Cold) aquaculture

1. **Carps-** Carps form the mainstay of aquaculture practices in India, contributing over 85% of the total aquaculture production. **India is called the “Carp Country” since carps** have been cultured since ancient times and been a relished delicacy in the country.

The carps which are native to the Indus-Ganges River Systems/ Indo-Gangetic Plains of India are referred to as the **Gangetic Carps / Indian Major Carps (IMC), comprising of Catla, Rohu and Mrigal** that contribute 60% of total Carps production.



Figure 5 - Rohu



Figure 6- Labeo fimbriatus



**Figure
7 –
Mrigal**

The carps that were introduced from other countries are referred to as **Exotic Carps** such as **Silver Carp, Grass Carp and Common Carp**.

There are also smaller carps often referred to as **Minor Carps** such as **Reba (Cirrhinus reba), Bata (Labeo bata), Fringe-lipped carp (Labeo fimbriatus), Calbasu (Labeo calbasu), white carp (Cirrhinus cirrhosus) and Cauvery carp (Labeo kontius)**.

2. Catfish-

Catfishes are a diverse **group of ray-finned fish named for their prominent barbels**, which resemble a cat's whiskers (but not all catfish have prominent barbell). Although **catfish can generally be found in faster-flowing rivers and streams, some catfish species have adapted to living in shallow salt-water environments** while other catfish species live their lives in caves underground.



Figure 8 – Magur (walking cat fish)

Most catfish are bottom feeders as they are negatively buoyant (which means that they usually sink rather than float due to a reduced gas bladder and a heavy, bony head). The **air-breathing catfishes such as Magur and Singhi inhabit shallow** waters, withstand low oxygen conditions and are referred to as "live fishes"; they are marketed live and fetch higher price.

Ex Magur-Walking Catfish Clarias magur,

3. Featherbacks

Featherbacks are **adapted to flowing conditions and widely distributed in deep and clear waters in the rivers, beels, reservoirs and ponds**.

The Bronze Featherback is **reported to enter brackish water**.

They are **carnivorous and predatory fish and feed on aquatic insects, molluscs**, shrimps and small fishes and take insects and tender roots of aquatic plants during early stages of life.

They **are rich in nutritive value and commands higher market price despite** the presence of a large number of intramuscular spines.

Ex- Chital Chitala chitala

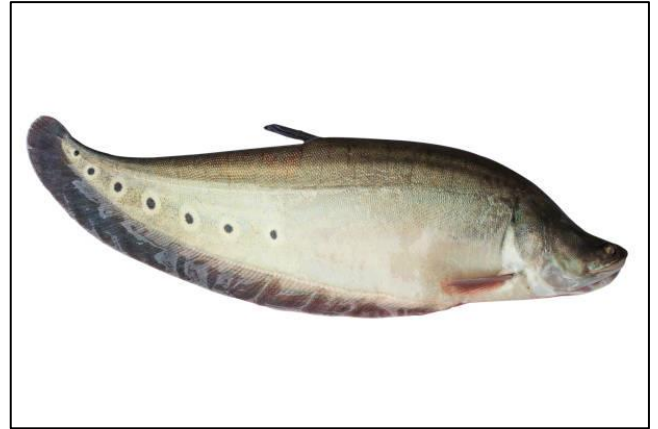


Figure 9 - Chital

4. Small Indigenous Fish Species (SIFS)

They are **defined as fish that grow to a maximum size of 25-30 cm in mature** or adult stage of their life cycle.

- They inhabit **rivers and tributaries, floodplains, ponds and tanks, lakes, beels, streams, lowland areas, wetlands and paddy fields.**
- In **India, out of 877 native freshwater fish species**, about 450 are Small Indigenous Fish Species (SIFS).
- **Some cultivable SIFS -are Mola, Climbing Perch, Barbs, Bata, etc.**



Figure 10 – Barbs (Barbus)

5. Snakeheads

They **inhabit swampy waters and their gills are adapted to breathe air.**

They can survive **out of water for up to four days**, provided they are wet, and are known to migrate up to **400 metres on wet land** to other bodies of water by wriggling with their body and fins.



Figure 11- Striped Murrel

The snakehead **meat has good taste, high nutrient** and also has high pharmaceutical values.

Example - Striped Murrel / Snakehead *Channa striata*, Spotted Murrel / Snakehead *Channa punctata*,

Coldwater Fishes

Coldwater fishes occupy an important place amongst the freshwater fishes of India. The **Coldwater fisheries deal with fisheries activity in water where temperature of water ranges from 5 to 20 degrees centigrade.**

The gills of cold-water fish are greatly reduced and the **gill opening are smaller in size for adaptation to cold temperatures.**



Figure 12 – Rainbow Trout

Important coldwater fishes of India are Mahseers such as *Tor putitora*, *T. tor*, *T.*

khudree, *T. mosal*, Snow Trout such as *Schizothorax richardsonii* and *Schizothoracichthys esocinus*, Mountain Trout such as *Barilius vagra*, *B. bendelisis*, other fish such as *Glyptothorax sp.*, *Garra sp.*, etc.

Example

1. **Golden Mahseer *Tor putitora***
2. **Tor Mahseer *Tor tor***
3. **Rainbow Trout *Oncorhynchus mykiss***
4. **Mountain Trout *Barilius vagra***



Figure 13 - Golden Mahseer

Brackish water Fisheries

Brackish water sources - coastal lagoons, lakes, backwaters, tidal creeks, canals, mudflats, mangrove plants, etc.

These water **bodies** lying between the freshwater and marine regimes have certain **characteristics**:

- **Fluctuating water level** synchronizing with the tides,
- Wide salinity **range of 0-35 ppt**,
- Higher **nutrient content** and productivity,
- **Serve as nursery grounds** for numerous marine organisms,
- Harbour a **rich diversity** of flora and fauna,
- Support artisanal **capture fisheries and provide livelihood** to the coastal fishers.

The tidal currents carrying with them the juvenile fish and shrimp, and the rising and falling water level formed the basis of traditional brackish water **fish and shrimp farming practices in the Bheries of West Bengal and Pokalli paddy fields of Kerala**.

The most **commonly cultured fish** are Mullet, Milkfish, Seabass, Pompano, Grouper, etc.

1. Mullet-

Thirteen species of Mullet are found in India of which eight species contribute to the commercial catches. **Mugil**

cephalus and M. parsia are the two cultivable species. They are **euryhaline species that tolerate salinity from 0-35 ppt**. They are **omnivorous** in feeding habit



Example

- Grey Mullet Mugil cephalus
- Goldspot Mullet Mugil parsia

Figure 14 - Mugil cephalus

2. Milkfish-

The **Milkfish, *Chanos chanos***, is the sole living representative of the family **Chanidae** and is widely distributed in the whole of tropical and subtropical regions of the Indian and the Pacific Oceans. The distribution coincides with coral reef areas where the water is



Figure 15 - Milkfish (*Chanos chanos*)

warm, clear and shallow. It is one of the most important species cultured in South-East Asia.

3. Bass-

Asian Seabass, *Lates calcarifer*, commonly called the **Giant Sea Perch** is an economically important food fish in the tropical and subtropical regions of Asia and the Pacific. Because of its **relatively high market value**, it has become an attractive commodity of **both large and small-scale aquaculture enterprises**. The major constraint to rapid expansion of Seabass culture has been the inconsistent supply of fry collected from the wild. The



Figure 16 - *Terapon jarbua*

Crescent Bass or Tiger Bass, *Terapon jarbua*, is common species in the Indo-Pacific; it occasionally makes its way into the aquarium trade.

Example

- Asian Seabass *Lates calcarifer*

- Tiger Bass *Terapon jarbua*

4. Pearlspot-

The Green Chromide **Pearlspot** (*Eetroplus suratensis*) is the state fish of Kerala and locally (in Malayalam) it is called “Karimeen”.

It being an **euryhaline species** is **compatible for polyculture with other brackishwater / freshwater fish**.

It **breeds naturally in confined conditions**; sexes are separate and fertilization is external.



Figure 17 - Pearlspot (*Etroplus suratensis*)

5. Grouper-

High consumer demand, desirable taste, hardiness in a crowded environment, efficient feed conversion, and rapid growth.

Commonly found species in Indian waters are **Greasy Grouper** (*Epinephelus tauvina*), **Orange-spotted Grouper** (*E. coioides*), **Brown-lined Reef Cod** (*E. undulosus*) and **Brown-spotted Reef Cod** (*E. chlorostigma*).



Figure 18 - Orange-spotted Grouper (*E. coioides*)

6. Scat-

Spotted Scat (Scatophagus argus) is a euryhaline subtropical fish widely distributed in Indian-Pacific waters. It occurs in two basic colour morphs which are called Green Scat and Ruby or Red Scat and lives in coastal muddy areas, including estuaries, mangroves, harbours, and the lower courses of rivers.

It is an important aquaculture food fish with high economic value and is also a popular aquarium species due to its colourful appearance, hardiness, slow growth, and calm behaviour.



Figure 19 - Spotted Scat (Scatophagus argus)

7. Pompano-

The **Silver Pompano (Trachinotus blochii)** belongs to the family Carangidae; it resembles the much sought after Silver Pomfret (Pampus argenteus). It is a tropical species, distributed in the Indo-Pacific Region, inhabits shallow coastal waters. It is one of the most promising species for coastal aquaculture as its growth rate is high, meat quality is good and fetches high price in the market.



Figure 20 - Silver Pompano (Trachinotus blochii)

Marine Fisheries

The Marine Fisheries has been playing a pivotal role in the Indian Fisheries Sector. The major marine catches come from the coast of Gujarat, Tamil Nadu, Kerala, West Bengal and Maharashtra.

The important Marine Fisheries can be grouped into the following categories:

- 1. Surface-water Fish (Pelagic):** Sardines, Anchovies, Ribbonfish, Mackerel, Seerfish, Tuna, etc.
- 2. Mid-water Fish (Pelagic):** Bombay Duck, Cobia, Silver Bellies, Horse Mackerel, etc.
- 3. Bottom-water Fish (Demersal):** Perches, Catfish, Pomfrets, Flatfish, Eels, etc

1. Sardines-

Sardines are **one of the two most important commercial pelagic schooling fishes** in India (the other being Mackerel). The **Oil Sardine *Sardinella longiceps*** alone contributes to the **largest single species fishery in the country**, comprising about 15% of the total marine fish catches.



Figure 21 - Oil Sardine *Sardinella longiceps*

They are **known to inhabit mostly the near-shore waters up to 25-30 m depth zone**. **Four regions of Lesser Sardine concentrations are noticed along the Indian coasts**, viz., Goa, Karnataka, Kerala, Tamil Nadu, Pondicherry, Andhra Pradesh and Orissa.

2. Mackerels

Mackerel is a common name applied to a number of different species of schooling **epipelagic fish of the family Scombridae**. They are found in **both temperate and tropical seas, mostly living along the coast** or offshore in the oceanic environment.



Figure 22 - Indian Chub Mackerel (*Scomber indicus*)

Mackerels found in Indian waters are **the Indian**

Mackerel (*Rastrelliger kanagurta*) and the Indian Chub Mackerel (*Scomber indicus*).

3. Anchovies-

Anchovies are **small fish having greenish-blue reflections** due to a silver-coloured longitudinal stripe that runs from the head to base of caudal (tail) fin.

There are **two fishing seasons: January to May and September to November**, the latter period being the peak season for anchovy catches.



Figure 23- Anchovies

4. Seerfishes

Another group of commercially important fishes, belonging to the family Scombridae, are the **Seerfishes represented in Indian coastal waters by Narrow-barred Spanish Mackerel** *Scomberomorus commerson*

5. Tunas-

Tunas (family Scombridae) are among the largest, most specialized and commercially important of all fishes. They are **found in temperate and tropical oceans** around the world and account for a major proportion of the world fishery production. **Tunas are unique among fishes because they possess body temperature several degrees higher than the ambient waters and have high metabolic rate that enables** them to exhibit extraordinary growth rate.



Figure 24 - Oriental Bonito (*Sarda orientalis*)

Example-Oriental Bonito (*Sarda orientalis*), Yellow-fin Tuna (*Thunnus albacares*)

6. Bombay-duck –

Bombay-duck (*Harpodon nehereus*) **found in the mid-water column**, distributed in the Indo West Pacific, is a **white, soft and delicate fish, predatory in habit**; it contributes to a major fishery along the **Maharashtra and Gujarat coasts** and occurs sporadically along the east coast.

It is captured using a **characteristic bag-net known as ‘dol’** that is **operated against tidal currents**. It is a **popular food-fish along the north-west coast of India**; it is marketed fresh, dried or salted.



Figure 25 – Bombay Duck

7. Cobia-

Cobia, *Rachycentron canadum*, is found throughout the **water column** and occurs in warm temperate to tropical waters of the West and East Atlantic, throughout the Caribbean and in similar zones in **the Indo-West Pacific region of India, Australia and Japan**. The fish is a **carnivore and congregates at reefs, ship wrecks, harbour-buoys** and other submerged structures in search of food.



Figure 26 - Cobia, *Rachycentron canadum*

- Juveniles may enter estuaries and mangroves in search of prey.
- It is **dark brown in colour**, young ones have **alternate black and white horizontal stripes**, it **grows up to 2 m length and 50 kg weight**.
- It is **one of the fast-growing and high-value marine fish that is being farmed in Sea Cages, in India**.

8. Flatfishes

Flatfishes (Order Pleuronectiformes) are bottom dwelling deep sea bony fishes with oval shaped and flattened bodies having both eyes on one side of the body. Flatfish can camouflage themselves by imitating the colour of the sea floor.

They include Halibut, Flounder, Sole, Turbot, Plaice, Dab, etc. Among the marine fishery resources of India, the flatfishes occupy a place **importance along the West Coast especially the North Kerala and South Kannada coasts.**



Figure 27 - Flatfishes (Order Pleuronectiformes)

9. Shellfish Fisheries

Shellfish are aquatic invertebrates having an exoskeleton/shell. **They can be grouped into two general categories: Crustaceans and Molluscs.** Crustaceans refers to those organisms that have **segmented bodies covered with armour-like sections of thick or thin shell** and bear jointed appendages: e.g., crabs, shrimps, lobsters and crayfish.

Molluscs are of three types:

- (i) Bivalves that **include clams, oysters, mussels** and scallops
- (ii) **Univalves that include abalone, periwinkle** and other snails like conch and whelk
- (iii) **Cephalopods that have a soft pliable body with an internal shell**, a beaked head bearing tentacles, include squids and octopus.



Figure 28 – Types Of Shellfish

Types of Shellfish

- **Crustaceans:** Prawns, Shrimps, Crabs, Lobsters, Crayfishes, etc.
- **Molluscs:** Gastropods (snails), Pelecypods or Bivalves (clams, mussels, oysters) and Cephalopods (squids, cuttlefish, octopus).

Crustaceans- Crustaceans are a large and diverse group of Arthropods (bearing jointed appendages) that thrive in a wide range of habitats; majority are marine while relatively few inhabit inland waters.

1. Shrimps and Prawns-

Shrimps and Prawns belong to the Order Decapoda (having 10 pairs of legs: 5 pairs of walking legs and 5 pairs of swimming legs). Generally, **the smaller forms are termed Shrimp while the larger ones are known as Prawns.**



Figure 29 – Shrimp And Prawns

Freshwater Prawns- The second largest freshwater palaemonid prawn, **Macrobrachium malcolmsonii**, also called the **Indian River Prawn/Monsoon River Prawn/Godavari Prawn**, has great potential for aquaculture development in the inland waters of the country.

Marine Shrimp- Aquaculture purpose, the exotic **Pacific White Shrimp** (*Litopenaeus vannamei*) has been introduced into India and is being farmed extensively on a commercial-scale in **low-saline or brackish water ponds in Gujarat, Karnataka, Kerala,**

Tamil Nadu, Andhra Pradesh, Odisha, West Bengal etc. With the availability of **Specific Pathogen Free (SPF) quality brooders and seed and adoption** of improved culture techniques, mortality of juvenile and young shrimp due to disease reduced and thereby yields of shrimp increased significantly in the country.

2. Crabs

Crabs inhabit a wide variety of **habitats like mud flats, rock crevices, under stones, in gravel and sand and constructed burrows** etc. They are primarily nocturnal and prefer hiding places for shelter during the day time.

Freshwater Crabs- They are generally **omnivorous, feeding on plant material**, live or dead animals such as fish, prawns, molluscs, etc., and sometimes cannibalism is also seen. **Indian Freshwater Crab Barytelphusa cunicularis**,

Brackishwater/Marine Crab- Commercial-scale crab culture is undertaken along the coastal areas of Andhra Pradesh, Tamil Nadu, Kerala and Karnataka.

Grow-out culture ('Crab Fattening') is generally pond based wherein crabs' juveniles are held in pens or cages. Production of **'Soft-Shelled Crabs'** is done using floating boxes held in tidal ponds.

The crab fishery in India is dominated by a few species such as Scylla serrata (Mud Crab), Portunus pelagicus, P. sanguinolentus (Three-spot Swimming Crab), Charybdis feriata

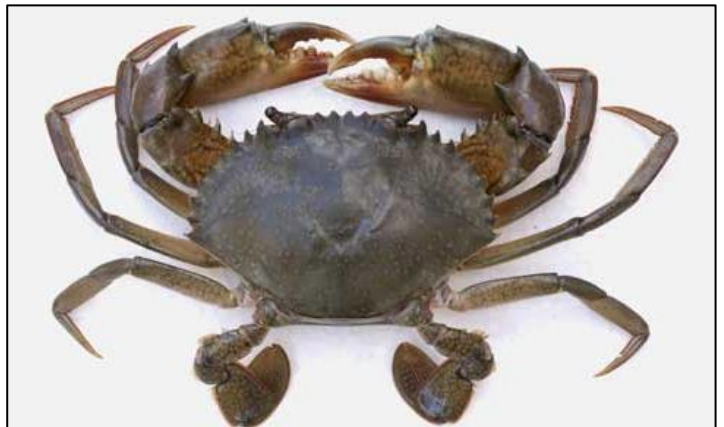


Figure 30 - Scylla serrata (Mud Crab)

3. Lobsters

Lobsters are one of the most valuable and highly priced crustaceans as well as an important export commodity. Lobsters are **solitary animals that crawl on the sea floor and hide under rocks**.

➤ Among the **12 species recorded from Indian coastal waters**, only **four species of Spiny Lobsters (Panulirus homarus, P. ornatus, P. polyphagus, P. versicolor)** and one species of Slipper Lobster (Thenus orinetalis) are commercially important.

- **Growth is slow, most lobsters take more than 3 years to attain maturation** and they have a prolonged life-cycle.
- **Grow-out and fattening of lobsters is commonly done in indoor tanks** and cages to fetch better price in market.

Figure 31 - Panulirus Homarus (lobster)



4. Molluscs

Molluscs are invertebrates, belonging to one of the largest phylum, inhabiting the marine, freshwater and terrestrial ecosystems. Their body structure, form, colour and size vary widely. **In India, molluscs are captured for food, pharmaceuticals and ornamental purpose and dead shells are collected for making shell-lime. Molluscs include forms such as bivalves, univalves and cephalopods, and some of them support both capture and culture fisheries.**

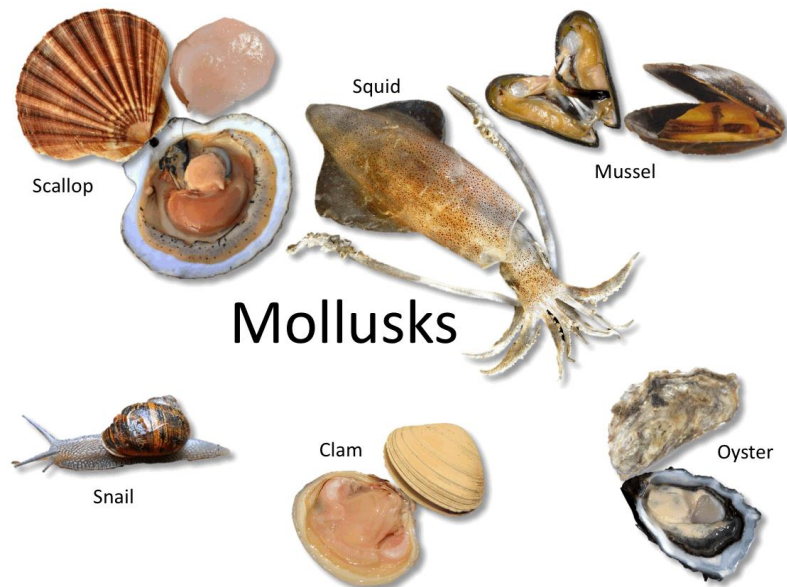


Figure 32 – Types Of Molluscs

Bivalves

Bivalves as the **name suggest are bilaterally symmetrical, laterally compressed molluscs**, with extensive mantle lobes, which secrete the shell having two valves. Majority of bivalves are marine and **a few species are found in freshwater habitats.**

Nearly 652 species of marine bivalves are reported from India, of **which 88 species are endemic to Indian waters.** Clams are the most important group among **bivalves forming 85.8%, followed by Mussels 9.6% and Oysters 4.6%.**

Mussels

Commercial fishery of Mussels along the Indian coast is mainly for the **Green Mussel *Perna viridis***, contributing **83.7%** on an average, and the remaining by the **Brown Mussel *Perna indica*** which is limited to the fishery along southern tip of Indian peninsula.



Figure 32 - Green Mussel *Perna viridis*

Oyster-

The **Indian Backwater Oyster *Crassostrea madrasensis*** is the most important edible oyster exploited (90.1%), followed by the Rock Oyster *Saccostrea cucullata* (5.9%) and Windowpane Oyster *Placuna placuna* (3.6%) along the Indian Coast.

The Pearl Oyster *Pinctata fucata*, which produces Natural Pearls in the sea, is utilized for production of Cultured Pearls.

Indian Backwater Oyster *Crassostrea madrasensis*



Figure 33 - Indian Backwater Oyster (*Crassostrea madrasensis*)

Ornamental Fish culture

Ornamental fishes are no longer an entertainment for the hobbyist alone, they are now a craze the world over in the form of a big business. The world trade in ornamental fishes has touched a record figure of over US\$ 1 billion that is growing at about 10% per year with Asia sharing almost 70% of the total export trade a large number of freshwater ornamental fishes from India that belong to the family



Figure 34 - Zebra Danios

Cyprinidae are already known to the world. The family includes the colourful barbs and loaches which are abundant in almost all rivers and hill streams. *Puntius*, *Brachydanio*, *Danio* well-known genera from the family Cyprinidae. These besides, there are certain catfishes which are equally popular.

Stages In the Life Cycle of Fish

Fish life cycles **vary among species**.

In general, however, fish progress through the following life cycle stages:

General life cycle of a fish

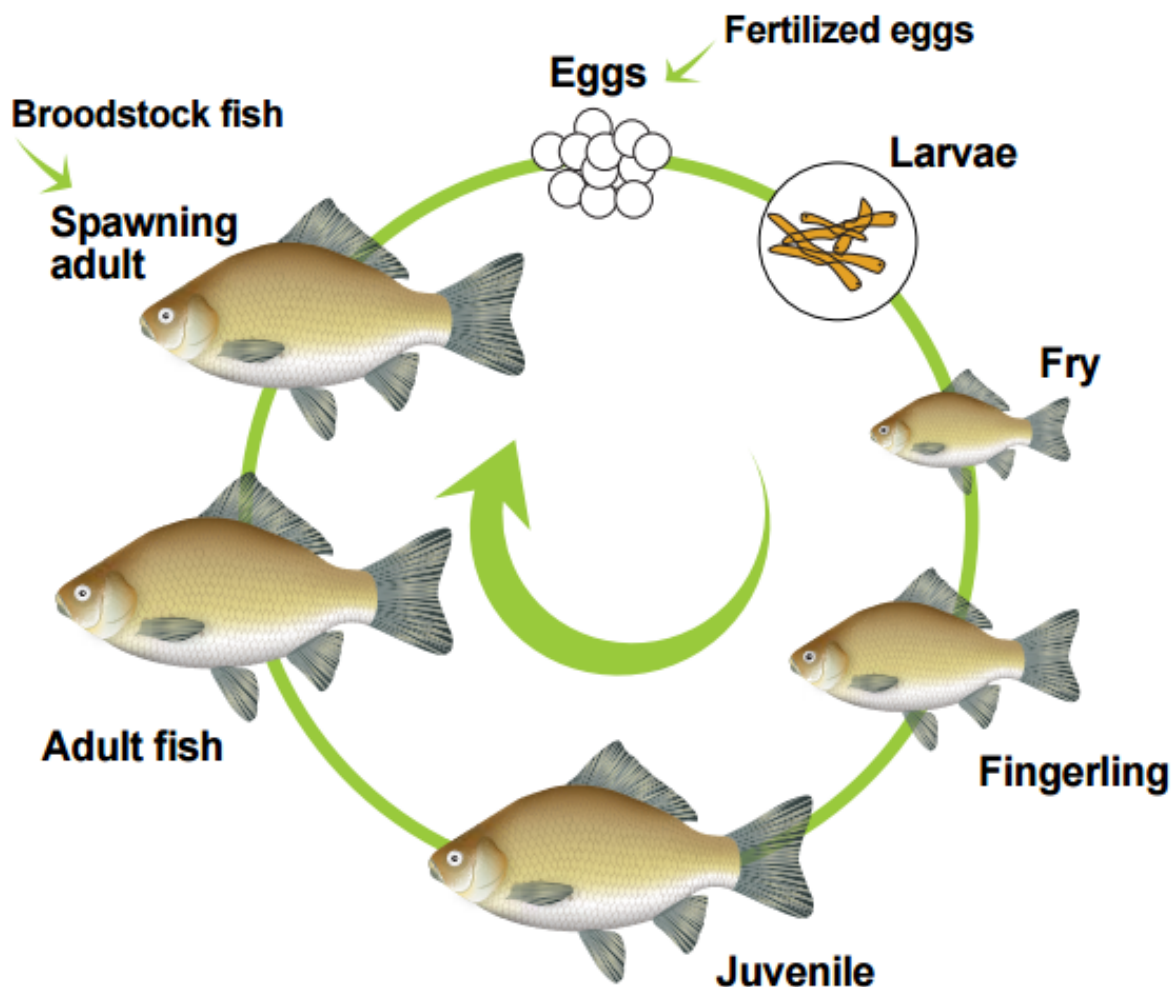


Figure 35 – Life cycle of fish

1. Eggs:

Fertilized **eggs develop into fish**. Most eggs do not survive to maturity even under the best conditions. **Threats to eggs include changes in water temperature and oxygen levels**, flooding or sedimentation, predators and disease.

2. Larval:

Larval **fish live off a yolk sac attached** to their bodies. When the yolk sac is fully absorbed, the **young fish are called fry**.

3. Fry:

Fry are ready to **start eating on their own**. Fry undergo several more developmental stages, **which vary by species, as they mature into adults**. Young fish are generally **considered fry during their first few months** (during their first few months to just less than one year in some species).

4. Juvenile:

The time fish spend **developing from fry into reproductively mature adults varies among** species. **Most fish do not survive to become adults**. Threats to survival include fluctuations in water temperature, changes in oxygen levels, competition for habitat and predators.

5. Adult:

When **fish are able to reproduce, they are considered adults**. The time it takes to reach maturity varies among species and individual fish. Fish with shorter life spans reach maturity faster. **For example, female round gobies mature in approximately one year and live for two to three years**. Lake sturgeon can live from 80-150 years, but females don't reach maturity until they are **approximately 25 years old**.

6. Spawning:

Female **fish release eggs into the water** (either into the water column or into a nest) and male fish fertilize **eggs by releasing milt**. Not all eggs are fertilized. Some fish spawn each year after reaching maturity, others spawn at intervals (**every four years, for example**), **while others spawn only once and then die**.

System Of Aquaculture

Aquaculture i.e. the **farming of aquatic organisms (animals & plants)** has come a long way and has developed into a **multi-million industry employing millions** of people around the globe. This is due to the vastness of the aquatic resources and diversity

Some Of the Classifications Are Listed Below:

1. On The Basis Of Salinity –

- **Fresh water** - Culture of fish in freshwaters, **where the salinity is zero (less than 0.05 ppt)** Ex- **carp culture like Rohu, Catla, Mrigal, rainbow trout**, mahaseer, *Macrobrachium rosenbergii* (freshwater prawn) etc.
- **Brackish water farming**- Culture of fish species in the **salinity range of 0.5-30 ppt**. The brackish water is **considered to be a mixture of freshwater and marine waters** e.g. milk fish, sea bass, shrimps etc.
- **Marine water farming** **Marine water farming**: Farming in marine **waters where the salinity is above 30 ppt**. e.g. **sea weed culture, marine pearl and edible oyster culture, mollusc culture, grouper fish culture** etc.

2. On The Basis Of Intensity

- **Extensive**: Adoption of **traditional techniques of aquaculture i.e without any management practices** e.g. dependence on natural productivity and little control over the stocks. In the system the stock density is low hence low production.
- **Semi-intensive**: Adoption of **mid-level technology, partial fertilization, dependence on natural productivity, supplementary feeding, with stock manipulation**, medium level inputs and medium rate of production e.g. composite carp culture. This type of **systems has higher stocking densities** and higher productivity e.g. carp culture.

- **Intensive:** Adoption of full complement of culture **techniques including scientific pond design, fertilization, supplemental feeding or only feeding without fertilization**; full measure of stock manipulation, disease control, scientific harvesting, high level inputs and high rate of production. **These systems can be managed with very high stocking density to achieve any desired productivity. E.g. Trout culture.**

3. On The Basis Of Number Of Species

- **Monoculture:** Culture of **single species in** a system
e.g. rainbow trout culture.
- **Polyculture:** Culture of **more than one species together in the same water body or enclosure is called as poly culture**
e.g. culture is of grass carp, common carp & silver carp together. **Polyculture of fish is also synonyms with composite fish culture.**

The principle “**Complete utilization of the pond's productivity for maximization of fish yield, the aim in composite fish culture, is achieved by growing, compatible fish species with complementary feeding habits occupying different ecological niches in the pond so that they do not compete with each-other for food and space**”.

The objective of **raising healthy and economically viable fish crops is realised through appropriate manipulation of fish stock and pond ecology.**

Composite fish culture technology in brief involves, the eradication of aquatic weeds and predatory fishes, liming, application of fertilizers on the basis of pond soil and water quality, **stocking with 100 mm size fingerlings of Indian major carps-catla, rohu, mrigal, exotic carps, silver carp, grass carp and common carp in judicious combination and density; supplementary feeding and harvesting of fish at a suitable time.**

Composite fish culture system is conducted by adopting three types of combinations viz., culture of Indian major caps alone, culture of exotic carps alone, and culture of Indian and exotic

4. On The Basis Of Type Of Enclosure

Pond Culture:

The majority of aquaculture throughout the world is done in ponds. Earthen ponds or reinforced concrete ponds are used for **culturing the fish, shrimp, prawn, etc. in both freshwater and brackish waters. Ponds are man-made impoundments filled with water.** The ponds may **be filled with canal water, rain water, bore well water or from other water sources in which the aquatic animals** such the finfish and shellfish are reared. The **climate, topography, water availability, soil quality and many other factors of the region** are taken into consideration for selecting a site for the pond.



Figure 36 – Pond culture

Cage Culture:

Cage aquaculture **involves the growing of fishes in existing water resources while being kept in a net cage which allows free flow of water.** It is an aquaculture production **system made of a floating frame, net materials and mooring system (with rope, buoy, anchor etc.)** with a round or square shape floating net to hold and culture large number of fishes and can be installed in reservoir, river, lake or sea.

There are 4 types of fish-rearing cages namely:

- Fixed cages,
- Floating cages,
- Submerged cages
- Submersible cages.

Farming of fish in **an existing water body removes one of the biggest constraints of fish farming on land**, i.e., the need for a **constant flow of clean, oxygenated water.** Cage farms are positioned in such a way to utilize natural currents, which provide the fish with oxygen and other appropriate natural conditions.

Fish Culture in Cages Have Both Advantages and Disadvantages, Some of These Are

Advantages

- **Ponds, strip and existing water resources like**, including lakes, reservoirs, pits, streams and rivers can be used.
- A relatively **low initial investment**.
- **Harvesting is simplified**.
- **Observation and sampling of fish is simplified**.
- **Less manpower** requirement.
- **Additional income** to fishers during closed seasons



Figure 37 – Cage Culture

Disadvantages

- **Feed must be nutritionally complete** and kept fresh.
- **Low Dissolved Oxygen Syndrome (LODOS)** is an ever-present problem and may require mechanical aeration.
- **Fouling** of net cage.
- The incidence of **disease can be high and diseases may spread rapidly**.
- **Vandalism or poaching** is a potential problem.
- **Navigation issues**.
- **Accumulation of unused feed and excreta will lead** to water pollution as well as eutrophication.

Pen Culture:

Pens are the specially designed nylon or bamboo made enclosures constructed in a water body into which fish are released for culture. Such type of culture is referred to as pen culture.

Pens are similar to cages except that the bottom is natural and the water body bottom is used. They are transitional structure between ponds and cages formed by damming a bay, lake, estuary, river, lake or reservoir. Pens are those where barriers can be constructed, in order to reduce the costs and increase the ease of operation. The site selection for installation of 'pen' is very crucial and should be sheltered against winds, should have an operational depth of 1-1.5.

They are usually constructed from **natural materials like bamboos, wooden logs**; however, **nylon nets are also in vogue**.

The major advantages of pen culture include, it being **continuous process due to continuous supply of natural water, greater production is assured in a limited space with rich food and oxygen supply, more energy is saved for body building due to reduced energy used in locomotion and feeding** etc.



Figure 38 - Pen Culture

Raceway Aquaculture:

Raceways are elongated structures with lengths quite larger than their width, made usually of concrete. Raceways are generally **constructed in a ratio of 5 to 1 (or greater) length to width, and with a depth of 3 to 5 ft.**



Figure 39 - Raceway Aquaculture

- Water should flow **evenly through the system to eliminate areas of poor water circulation** where waste materials or sediment may accumulate.
- Raceways may be constructed above ground.
- Fish cultured in raceways **require a large quantity of good quality water, preferably supplied by gravity flow (trout culture).**
- On average, **1 to 3 gal per minute of flow should be available for each cubic foot of raceway volume at densities of 3 lbs. of fish per cubic foot.**

Recirculatory Aquaculture systems:

They are **closed or semi-closed systems**; a battery of ponds or raceways, **interconnected with each other in such a way that the same water is used over and over again with minimum wastage of water as outflow, with no more than 10% of water exchange.** The water is reconditioned **by clarification, biological filtration, and re-aeration so that**

most of the water is reused and only a fraction of the total daily flow is made up of new water.

The **productive capacity of this system depends on the filtration system's ability** to remove wastes, as well as on the volume of replacement water used to improve water quality.

Water quality **should be monitored frequently in such**

a system because without high **rates of water exchange, toxic metabolites may accumulate rapidly** if the biological filtration system is not sufficient to handle the wastes.

On The Basis Of Integration

Agriculture cum fish farming- This is generally referred to as “**integrated farming**”. In this type of **integrated farming fish culture is done in combination with selected agriculture crop (s) like paddy, vegetable, mulberry etc**, in such a way, that the waste products from one system act as an input for the other and vice versa.

This type of farming offers **great efficiency in resource utilization**, as waste or by product from one system is effectively recycled.

It also enables **effective utilization of available farming space for maximizing production**.

The rising cost of **protein-rich fish food and chemical fertilizers** as well as the general concern for

energy conservation has created awareness in the utilization of rice and other crop fields and livestock wastes for fish culture.



Figure 40 - Recirculatory Aquaculture systems



Figure 42 - Paddy Cum Fish Farming

Animal husbandry cum fish farming: This includes **poultry cum fish farming, duck cum fish farming, pig cum fish farming, cattle cum fish farming etc.** In all these systems, the **nitrogenous rich excreta** from these animal systems are used to replace the fertiliser and manure in the ponds, **hence not reducing only the cost of production but also help in reducing inorganic inputs like fertilisers** and provide two production systems to the farmer besides reducing the stress on the ecosystem.

Organic Fish Farming

The principle of organic aquaculture is to produce natural healthy food without having any adverse impact on the environment; this is done by **avoiding the use of chemicals, medicines, antibiotics, hormones etc in the production system.**

Organic aquaculture helps to maintain the sustainability of the system. The breeds used in **organic aquaculture should come from organic sources wherever available. Breeding should be done only using** local methods without using any inducing agents or other artificial breeding techniques.

Use of triploid, genetically modified species is avoided. The health management should be based on the concept of “Prevention is better than cure” in aquatic organism’s health management must be followed so that no need for medication exists.

Integrated Multi-Trophic Aquaculture System

It involves combining the farming of **multiple species from different trophic levels with complementary ecosystem function. Integrated multi-trophic aquaculture** borrows a concept from nature; namely, that in the food chain, one species always finds a feeding niche in the waste generated by another species. **IMTA involves cultivating organisms in a way that allows the uneaten feed, wastes, nutrients and by-products** of one species to be recaptured and converted into fertilizer, feed and energy for the growth of the other species. IMTA farmers combine species that need supplemental feed such as fish, with “extractive” species. **Extractive species include biofilters as, filter feeders (e.g., mussels) and deposit feeders (e.g., sea urchins), seaweeds (e.g., kelps), etc**

Biofloc Technology

BFT is an environment **friendly aquaculture technique based on in-situ microorganism** production. Biofloc Technology (BFT) is considered **as new “blue revolution”** wherein,

fish are **grown intensively (minimum of 300 g of biomass per square meter)** with zero or minimum water exchange and nutrients can be continuously recycled and reused in the culture medium.



Figure 43- Biofloc Culture

Biofloc is the suspended growth in ponds/tanks which is the aggregates of living and dead particulate organic matter,

phytoplankton, bacteria and grazers of the bacteria. It is the utilization of microbial processes within the **pond/tank itself to provide food resources for cultured organism** while at the same time acts as a water treatment remedy. Thus, this system is **also called as active suspension ponds or heterotrophic ponds or even green soup ponds.**

Management Principles Of Aquaculture

The management in **aquaculture is aimed to control the situation in order to prepare more fish.**

- With aquaculture management, **the predators can be controlled,**
- The **physico-chemical parameters of water can be kept under check,**
- Escape of the fish is prevented,
- **Health monitoring** and disease control and are provided with feed and other factors important for its growth and well-being.

All these aspects are **considered as aquaculture management practices** and are **primarily divided into two categories;**

1. Pre- Stocking Management Practices

2. Post Stocking Management.

Most of these practices are based on the concept of composite fish culture or polyculture in semi-intensive mode of fish farming in earthen ponds

For better management and ease of aquaculture operations, a composite fish farm ponds are categorised into three out/stocking ponds.

Nursery ponds:

- These are also **called transplantation ponds.**
- These are **seasonal ponds** and are constructed near the spawning and rearing ponds.
- The main **object is to create a suitable condition of food availability and growth of fry because at this stage they are most susceptible to hazards like the wave action and predators.**
- These should **be small and shallow ponds 0.02-0.06 ha.** in size and 1-1.5 m. in depth. In the nurseries, **the spawn (5-6 mm) is reared to fry stage (25-30 mm)** for about 15 days.

Rearing Ponds:

These should be **slightly larger but not proportionally deep.**

These should be **located near the nursery pond and their number may vary depending upon culture.**

They should **preferably be 0.08-0.10 ha in size and 1.5-2.0 m in depth. The fry (25 30 mm) are reared here upto the fingerling (100-150 mm) stage for about 3-4 months.**

Stocking Or Grow Out Ponds:

- These are the **largest ponds and are deeper, with a depth of about 2-2.5 m.**
- The size of the pond may **vary from 0.2-2.0 ha.,** but these should preferably be 0.4-0.5 ha in size. These are rectangular in shape.
- The fingerlings **and advance fingerlings are reared upto marketable size for about 6 months. One year old fishes may grow upto 1 kg. or more in weight.**

The pond management including pre-stocking & Post stocking management is generally applies to all the categories of ponds and is as follows;

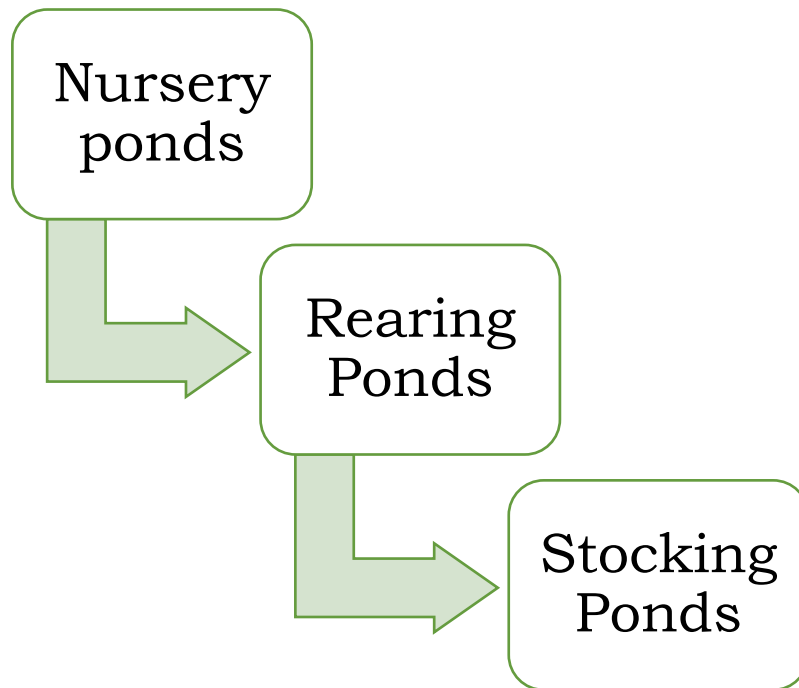


Figure 44 – Categorisation of Ponds

Pre-Stocking Management

It involves site selection, eradication of weeds, insects and predators, liming, manuring, etc.

Site selection- When considering **a site for the fish ponds**, several aspects have to be considered like the type and number of ponds to be constructed, the topography of the area, the water supply and the type of fishes to be reared.

Characteristics Of a Good Culture Pond.

- Ponds may be **dug into the ground**, they may be partly above and partly in the ground, or they may be below original ground level, **soil should contain a minimum of 25% clay**.
- **Depth should be 0.5-1.0 m** at shallow end, **sloping to 1.5-2.0 m** at the drain end
- Best shape for ponds is rectangular.
- Construct ponds **with 2:1 or 3:1 slope on all sides**.
- **Gate valves, baffle boards or tilt-over standpipes** should be provided; draining should take no more than 3 days
- Inflow lines should be of sufficient capacity to fill **each pond within 3 days**.

- Dykes should be sufficiently wide

After the construction of the pond at the selected site, the real management of ponds starts and the ponds need to be prepared such that the pond environment provides optimum condition for the growth of fish. The **steps involved in pre-stocking and post-stocking management are similar in the nursery, rearing and grow-out ponds; however, an additional step in the pre-stocking management in nursery ponds is the eradication of aquatic insects which predate on spawn and fry.**

1. Draining and drying-

Pond **needs to be drained and dried before culture operations begin** Drying facilitates **in oxidation of organic matter, removal of toxic gases such as ammonia and hydrogen sulphide**

- Its aides in killing pathogenic microorganisms, predatory fishes, weed fish and unwanted aquatic plants.
- A **drying period of 7-10 days** is sufficient in the ponds with clayey soil ponds.

2. Ploughing-

Ploughing of **dried ponds is usually carried out to mix up the of the soil** for oxidation of organic matter and mineralisation of the nutrients.

3. Liming-

Liming helps in **improving the soil texture, water retention, pH correction, disinfection of soil and water and nutrients like organic carbon**, available nitrogen, available phosphorous which help in enhancing the productivity of the ponds.

- It also helps **in pH correction** and bringing it to **the desirable range of 6.5 – 7.00**.
- The chiefly used lime sources in ponds are **agricultural lime or calcite (CaCO₃), dolomite [CaMg(CO₃)]**, Calcium hydroxide/slaked lime Ca(OH)₂ and Calcium oxide/quicklime – CaO.
- **Quick lime is preferred for applying to** soil and calcite agricultural lime for application to water after stocking of the ponds.
- Ex a soil with pH 4 requires about 2000 kg/ha of lime.

4. Pond fertilization/ Manuring –

The **successful fry rearing in ponds depends on the availability of zooplankton**. For sustained zooplankton production availability of phytoplankton and **bacterial base are important factors** **Phytoplankton production requires adequate nitrogen and phosphorus**.

- **Manuring increases natural productivity**, production of fish food organisms and in return increases fish production.
- Both organic manure and chemical fertilizers can be used. But depending on the economic viability and availability, organic manure is most preferred.
- **Organic manure such as cow dung, poultry dropping should be applied 7 days after the application of lime** and should be in dried form or in the form of slurry.
- Apply cow dung @ 1000kg/ha. This organic manure application **should be done 15 days before stocking of fish seeds**.
- **Initial manuring with organic manure @ 20% of the total requirement is done 15 days prior to stocking**.
- The remaining **80% of the organic manure may be applied in split** up 11 equal monthly doses during the rearing period.
- **Generally, urea or ammonium sulphate is used as a source of N while single or triple super phosphate is used as a source of phosphorus**. Their action is quick as compared to organic manures, not more than 200kg/ha is recommended if necessary

5. Aquatic Weed Control-

Aquatic weeds can **cause severe problems in fish farming** if allowed to grow in the ponds uncontrolled;

- they can **Compete for nutrients** with phytoplankton
- **Prevent light penetration** and suppress the production of phytoplankton
- **Cause oxygen super saturation** during day and oxygen deficiency during night
- Harbour aquatic **insects and predatory fish**

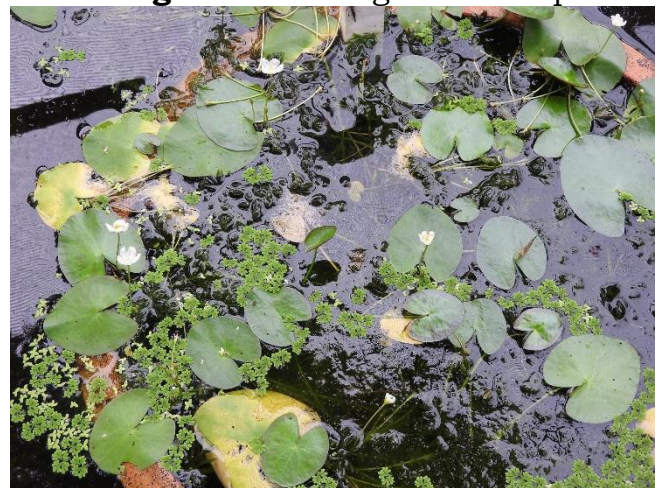


Figure 45 – Aquatic Weed Control

- Hinder free movement of **fish and reduce** their living space
- Cause problems **during harvesting by hindering netting** operations.
- Increase siltation in the **pond, reducing pond depth over a period of time.**

So, **their eradication is must to avoid the above problems**, this can be done **through, manual, mechanical, chemical and biological methods.**

Manual method- which involves physical removal of weeds by manual labour

Mechanical method - This method is **also convenient for small water bodies. For clearance of large areas by mechanical means, several types of machines have also** been evolved but their efficacies and economics have to be carefully assessed before making investment for such machines.

Chemical method: This method is **effective and long lasting in larger** water bodies. The weeds killed by **chemicals are allowed to settle at the bottom and ultimately recycled to become manure for enhancing fertility of fishery waters.** Sometimes, however, the sudden death and decomposition of large masses of vegetation may deplete dissolved oxygen in water causing even fish mortality

Biological method -**Grass carp (Ctenopharyngodon idella) is a very useful fish for biological control of some species of aquatic vegetation.** Many species of submerged vegetation viz. Hydrilla, Najas, Ceratophyllum etc. can be effectively controlled by stocking the weed infested waters with grass carps of 200 mm and above in size. **Algal bloom especially microcystis can be controlled by diuron @ 4 kg/ha-m water area.**

6. Control of Predatory and Weed fish

Predatory fishes **attack and eat the fry and smaller fishes in nursery** and rearing ponds and also injure bigger fishes causing mortality.

- Weed fishes on the other hand compete with **stocked fish for food, space and oxygen** and **result in reduction in production** of desirable fish.
- **Examples of Common predatory fish include murrels** (Snakeheads), Catfishes such as Wallago attu, Clarias batrachus, Heteropneustis fossilis, Ompok sp. Etc.,
- **The Weed fish include Puntius, Barbas Danio, Aplocheilus, Anabas etc.,** Dewatering followed by sun drying is most effective way to control weed and predatory fish.

- However, the ponds which cannot be dried, piscicides are used **ex - Derris root powder, Mahua oilcake**

7. Stocking Management-

To ensure that the **water is no longer toxic**, a simple test can be performed by releasing or **keeping few fingerlings in a 'hapa'**. If the **fingerlings survive for 24 hours**, the pond is considered fit for stocking.

Stocking density:

The pond will be ready for **stocking after 15 days of application of fertilizers**.

Fish fingerlings of 50- 100 gm size (approx.) should be **used for stocking @ 5000-8000 no's. per hectare**. The present model envisages stocking of advanced fingerlings and **rearing for 10-12 months**. When the rearing period is **limited to about 8 to 10 months**, a density of **6,000 fingerlings/ha** is most suitable

Rearing ponds are stocked with **carp fry @ 20 to 30 numbers per square meter** in rearing ponds while **carp grow out ponds** are stocked **@ 5000-1000 numbers per hectare** when polyculture is followed.

Post Stocking Management

Post stocking management involves monitoring and management of supplementary feeding, disease management, pond water quality management and harvesting the stock.

1. Nutrient Requirement-

Each **fish species has specific nutritional requirement** i.e. requirement for protein, energy, vitamins, minerals etc. which are essential for growth, reproduction and other normal physiological functions.

A deficiency of one or more of essential nutrients resulted in a reduced performance, occurrences of disease or even death.

These nutrients **may come from artificial feed or natural food** **Protein levels of 30-36%** have been found to be adequate and recommended in carp culture.

Depending on the **quality and quantity of dietary protein, the crude fat requirement of carp feed is 8% (spawn and fry) and 6% (grow out and brood stock) carp feed. Commercial grade vitamin premix and mineral premix can be supplemented in the fish diet @ 1-2% level and 2-4% respectively.**

Feeding Methods

In general, **good palatable feeds are consumed within 15 to 30 minutes by most species.**

- **Broadcasting the feeds** from pond bunds is **quite effective for small ponds (< 0.5 ha). But in larger ponds (> 0.5ha),** broadcasting the feeds from bunds should be supplemented by distribution in slightly interior areas using small boats.
- **Moist feeds and meals should not be broadcasted** but kept on earthen or plastic trays, placed in the peripheral areas of the pond bottom.

Feeding Practices for Carps

In rearing pond-

For feeding fry in well prepared **rearing pond for 90 days period, initially crumbles of 0.5 mm diameter can be used and gradually crumble size can be increased up to 0.8-1 mm dia.** Considering the plenty of live food available for larvae, daily allowance of **feed in rearing pond is 6-8% of body weight during first month, 5-6% of body weight during second month** Tray (plastic or aluminium) system of feeding method is always preferable for controlling the feed loss

In grow out pond-

During 10-12 months culture period, for feeding fingerlings and growers in grow out pond under composite fish culture system different types of feed such as dry mash, wet ball, cooked paste, cooked ball or dry pellets can be used.

Fish are **fed 5% of body weight during first two months, 4% of body weight during next two months, 3% of body weight during next two months, 2 % of body weight during next two months and 1% of body weight during final 2- 4 months** depending upon the water temperature. **The farm-made mash feed is comprised of rice bran and oil cake (groundnut, mustard and cottonseed cakes are widely used), typically in the ratio of 50:50.**

Supplementary Feeding

Supplementary feeding is provided to the stocked fish. At high stocking densities natural food produced through fertilization will not be sufficient. **Supplementary feeds could be simple mixtures of rice bran and oil cake in a ratio of 1:1. Spawn are fed at a rate of 8-10% of the biomass, while fry when stocked in rearing ponds up to fingerlings stage are generally fed @ 6-8% of the standing crop.**

In grow-out ponds the fish are fed initially at a rate of 5% which is gradually decreased to 2-3% biomass till harvest. **Periodic sampling of the stock is necessary** to estimate the biomass and adjust feeding rates.

Supplementary Fertilization-

To maintain the pond **Phyto and zooplankton production, reduced due to overfeeding by fish, the supplementary feed post-stocking may be required in order to maintain this productivity**, the ponds are subsequently **fertilized with cow dung at a dose of 0.5 tons/ha at fortnightly intervals.** However, supplementary fertilization should be followed with caution since **application of excess of fertilizers and manures may lead to poor water quality of the pond, or undesirable blue green algae such as Microcystis** may develop which apart from being toxic, will not constitute as food item for fish.

Health Management-

Disease is **a prime agent affecting fish mortality, especially when fish are young.** **Pathogens which can cause fish diseases comprise:** viral infections, bacterial infections, fungal infections, protozoan infections, water mould infection, etc. However, if any case of disease or infection is reported, the prophylaxis and treatment is done either by

- Disinfection of tanks and ponds
- Disinfection of culture units and equipment
- Providing water sources free of pathogens
- Optimisation of environmental conditions
- Therapeutic baths for fish

Water Quality Management

Good water quality **will ensure optimal environment for producing healthy fish** and should be maintained at all stages of production.

The physico-chemical **parameters of pond water should be within optimum limits for better fish growth and survival.** The dissolved oxygen is an important factor to be monitored. **Dissolved oxygen**, the source of which is from the atmosphere and also through photosynthesis by plankton and macrophytes, **should be ideally above 5 mg/l.** **Water temperatures ranging from 24-30°C is conducive to fish growth and lower temperatures below 18°C adversely affect growth.** The stocking period may thus be so manipulated that low temperature months are not included in the culture period.

The favourable ranges in water quality parameters for freshwater carp culture is presented in the following Table:

Parameters	Optimum limit for culture
Dissolved oxygen	Greater than 4ppm
Temperature	24-30°C
Un ionized Ammonia	Less than 0.05 ppm
Turbidity	30-45cm
Salinity	Less than 0.5ppt
Nitrite	Less than 0.1ppm
Nitrate	50-150ppm
Carbon dioxide	Less than 8ppm
Iron	Less than 0.5ppm
Total alkalinity	20-150ppm
Total hardness	20-200ppm
Hydrogen sulphide	Less than 0.002ppm

Important Facts

Slightly alkaline water is considered good for fish culture; with optimal range of 6.5-7.5. **Less than 5 and more than 10 pH is lethal to fish and prawns, known as acid death point and alkaline death point,** respectively.

Nitrogen- The available nitrogen contents in **the range of 25 to 50 mg/100 g** are considered favourable for average production.

Phosphorous- The available soil phosphorus in the **range of 3 to 6 mg/100g** is considered desirable for average productive soil.

Organic carbon -Organic carbon content in the **range of 0.5 to 1.5%** is considered is ideal for average fish production.

Water Temperature - Indian major carps can thrive well in **the temperature range of 18-38°C**, and rainbow trout has an **optimum temperature range of 12-20°C**.

Turbidity Of Pond Water - **Turbidity is measured with the secchi disc.** If the secchi disc **disappears at 30-50 cm** the water is productive in nature. **If it is not visible at a depth less than 25 cm, a dissolved oxygen problem could arise** during the night. If it is more than 50 cm, the plankton produced is less in the pond water.

Dissolved oxygen- Dissolved oxygen is **one of the most critical parameters of water** which **directly affects fish growth and pond productivity**. The main source of DO in ponds is **due to diffusion of atmospheric oxygen in water and photosynthesis**. The optimum **dissolved oxygen is 5-8 ppm**.

If less than **5 ppm the growth rate decreases and less than 1ppm of dissolved oxygen** may result in death.

A minimum concentration of 5 mg/l is sufficient for warm water fishes, while 9 mg/l is required for cold water sps. (For short periods even if DO falls by 2-3 mg/l below the minimum levels fishes won't die).

Fish Migration

Many types of **fish migrate on a regular basis, on time scales ranging from daily to annual, and over distances ranging from a few meters to thousands of kilometres**. Fish usually migrate because of diet or reproductive needs, although in some cases the reason for migration remains unknown.

Classification Of Fishes on the basis of Migration

- 1. Potamodromous fish** migrate within fresh water only
- 2. Oceanodromous fish** migrate within salt water only
- 3. Diadromous fish** travel between salt and fresh water

4. **Anadromous** fish live in the ocean mostly, and breed in fresh water
5. **Catadromous** fish live in fresh water, and breed in the ocean
6. **Amphidromous** fish move between fresh and salt water during their life cycle, but not to breed
7. **Forage fish** - Forage fish often make great migrations between their spawning, feeding and nursery grounds.

Important Terminology Related to Fisheries

Fish husbandry – the breeding and rearing for culture and management.

Fish Hatchery – one form of a fish farm, managed with the intent of breeding and seed production of fish

Zooplankton: Minute animals in water, chiefly rotifers and crustaceans, that depend upon water movement to carry them about, having only weak capabilities for movement; important prey for young fish.

Fry – Development stage of fish immediately after the larvae stage, **at an age of less than a week**

Fingerling – Development stage of fish following the fry stage and **continuing into the first three to four months of life.**

Yearling – Development stage of fish following the **fingerling stage and lasting until approximately one year of age.**



Figure 46 - Fry



Figure 47 - Fingerling



Figure 48 – Yearling

Brood stock – fish of any particular species which are raised for reproduction and seed production purposes.

Dissolved Oxygen (DO) – Level of oxygen in water and available for living organisms.

pH – A measure of the concentration of hydrogen ions present in a body of water indicating acidity. Measured **on a scale from 1 to 14; 7 is neutral**.

Raceway – Culture system which typically uses gravity to move water through system. The inlet is located higher than the outlet and the base of these typically rectangular units that slopes downward, supplying a rapid flow of water.

Water Quality – The physic-chemical condition of water as measured by a number of parameters including pH, ammonia, alkalinity, etc.

Turbidity – A measure of the clarity of water. Turbidity is closely related to total suspended solids and organic particulates.

Stocking- Fish stocking is the practice of **raising fish seed/fry/fingerlings in a hatchery** and releasing them into a pond or raceway (aquaculture system)

Weed fish: An unwanted fish species generally small fishes which are not the target of the producer and grow on their own in the system.

Biochemical oxygen demand (BOD): The amount of oxygen consumed by respiratory processes measured in a bottle of water incubated in the dark at constant temperature for a specific time. Standard BOD determinators are for 5-day incubations at 20°C. It should always be stated whether a water sample had been filtered (soluble BOD) or not (total BOD). The biochemical oxygen demand is often wrongly reported as biological oxygen demand.

Fish meal: Fish and fish-processing offal dried, often after cooking and pressing (for fatty fish), and ground to give a dry, easily stored product that is a valuable ingredient of animal feeding stuffs.

Liming: Addition of lime to a water body to reduce its acidity and helps in buffering activity.

Organic aquaculture-Organic aquaculture is the farming of aquatic animals like shrimp, fishes, bivalves and aquatic plants without using antibiotics, chemicals, and fertilizers by

preserving the ecosystem and biodiversity. It helps in raising aquatic products in sustainable and pollution free manner.

Eurythermal: Species with wide range of temperature tolerance.

Euryhaline: Species with wide range of salinity tolerance.

Stenothermal: Species with narrow range of temperature tolerance.

Stenohaline: Species with narrow range of temperature tolerance

Trophic level: The position that an organism occupies in a food web.

Table – Rearing period of Different organism under aquaculture

Fin Fish/Non-Fin Fish	Rearing Period
Carp	Usually, fish over 500 g should be harvested every 3–4 months with simultaneous stocking with fingerlings
Fresh water prawns	The duration of culture varies from 6 to 12 months depending on the type of culture practice. Generally, in monoculture the culture period may be 6-8 months under monoculture and 8-12 months under polyculture.
Shrimp	The average culture period required is around 120-150 days
Water crabs	Rearing of water crabs for 3-4 weeks